# Kubernetes – 03

**Task 1: Apache Pod + ClusterIP Service**

**First create the Apache Pod:**

```
  GNU nano 8.7.1                                                  apache-pod.yml *
apiVersion: v1
kind: Pod
metadata:
  name: apache-pod
  labels:
    app: apache
spec:
  containers:
    - name: apache
      image: httpd
      ports:
        - containerPort: 80
```

```
root@master:~# kubectl get pods
NAME         READY   STATUS    RESTARTS   AGE
apache-pod   1/1     Running   0          25s
root@master:~#
```

**create the ClusterIP Service**

```
  GNU nano 8.7.1                                                  apache-service.yml *
apiVersion: v1
kind: Service
metadata:
  name: apache-service
spec:
  selector:
    app: apache
  ports:
    - port: 80
      targetPort: 80
  type: ClusterIP
```

```
root@master:~# nano apache-service.yml
root@master:~# kubectl create -f apache-service.yml
service/apache-service created
root@master:~# kubectl get svc
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
apache-service   ClusterIP   10.107.12.178   <none>        80/TCP    8s
kubernetes       ClusterIP   10.96.0.1       <none>        443/TCP   4m8s
root@master:~#
```

**And check whether the Service is connected to the Apache Pod:**

```
root@master:~# kubectl get endpoints apache-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME             ENDPOINTS           AGE
apache-service   192.168.171.10:80   42s
root@master:~#
```

- **`apache-service` → Service name**
- **`192.168.171.10` → Apache Pod IP**
- **`80` → Apache port**
- **The endpoint is present, so the Service is correctly selecting the Pod.**

**The warning about `Endpoints` being deprecated is not an error. Kubernetes is just recommending the newer EndpointSlice API.**

**Task 2: Expose an Nginx Pod externally using NodePort.**

**create the Nginx Pod:**

```
 GNU nano 8.7.1                                                  nginx-pod.yml *
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  labels:
    app: nginx
spec:
  containers:
    - name: nginx
      image: nginx
      ports:
        - containerPort: 80
```

```
root@master:~# nano nginx-pod.yml
root@master:~# kubectl create -f nginx-pod.yml
pod/nginx-pod created
root@master:~# kubectl get pods
NAME         READY   STATUS    RESTARTS   AGE
apache-pod   1/1     Running   0          4m45s
nginx-pod    1/1     Running   0          8s
root@master:~#
```

**Expose Nginx using NodePort**

```
 GNU nano 8.7.1                                                  nginx-nodeport.yml *
apiVersion: v1
kind: Service
metadata:
  name: nginx-nodeport
spec:
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
  type: NodePort
```

```
root@master:~# nano nginx-nodeport.yml
root@master:~# kubectl create -f nginx-nodeport.yml
service/nginx-nodeport created
root@master:~# kubectl get svc nginx-nodeport
NAME             TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
nginx-nodeport   NodePort   10.108.137.98   <none>        80:30080/TCP   10s
root@master:~# kubectl get endpoints nginx-nodeport
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME             ENDPOINTS           AGE
nginx-nodeport   192.168.37.199:80   18s
root@master:~# ^C
root@master:~# kubectl get svc nginx-nodeport
NAME             TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
nginx-nodeport   NodePort   10.108.137.98   <none>        80:30080/TCP   44s
root@master:~#
```

```
root@master:~# kubectl get svc nginx-nodeport
NAME             TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
nginx-nodeport   NodePort   10.108.137.98   <none>        80:30080/TCP   44s
root@master:~# ^C
root@master:~# kubectl get nodes -o wide
NAME         STATUS   ROLES           AGE   VERSION    INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION   CONTAINER-RUNTIME
master       Ready    control-plane   44h   v1.34.13   10.30.1.170   <none>        Ubuntu 26.04.1 LTS   7.0.0-1014-aws   containerd://2.2.2
worker-01    Ready    <none>          44h   v1.34.13   10.30.1.102   <none>        Ubuntu 26.04.1 LTS   7.0.0-1014-aws   containerd://2.2.2
worker-02    Ready    <none>          44h   v1.34.13   10.30.1.196   <none>        Ubuntu 26.04.1 LTS   7.0.0-1014-aws   containerd://2.2.2
root@master:~# ^C
root@master:~# curl http://10.30.1.102:30080
^C
root@master:~# curl http://10.30.1.102:30080
curl http://10.30.1.102:3008^C
root@master:~# curl http://10.30.1.102:30080
^C
root@master:~# kubectl get svc nginx-nodeport
kubectl get pods -o wide
NAME             TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
nginx-nodeport   NodePort   10.108.137.98   <none>        80:30080/TCP   8m58s
NAME        READY   STATUS    RESTARTS   AGE   IP              NODE        NOMINATED NODE   READINESS GATES
apache-pod  1/1     Running   0          15m   192.168.171.10  worker-01   <none>           <none>
nginx-pod   1/1     Running   0          10m   192.168.37.199  worker-02   <none>           <none>
root@master:~# ^C
root@master:~#
```

**Your Kubernetes configuration looks correct.** ✅

- **nginx-pod → Running**
- **Nginx Pod is on `worker-02`**
- **Service → NodePort**
- **Port mapping → `80:30080`**
- **Service selector is finding the Nginx Pod**

**So the NodePort itself is configured correctly.**

**Task 3: ReplicationController with 3 Nginx replicas**

```
  GNU nano 8.7.1                                                  nginx-rc.yml *
apiVersion: v1
kind: ReplicationController
metadata:
  name: nginx-rc
spec:
  replicas: 3
  selector:
    app: nginx-rc
  template:
    metadata:
      labels:
        app: nginx-rc
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

```
root@master:~# nano nginx-rc.yml
root@master:~# kubectl create -f nginx-rc.yml
replicationcontroller/nginx-rc created
root@master:~# kubectl get rc
NAME       DESIRED   CURRENT   READY   AGE
nginx-rc   3         3         3       6s
root@master:~#
```

```
root@master:~# kubectl get pods -l app=nginx-rc
NAME             READY   STATUS    RESTARTS   AGE
nginx-rc-4bt9c   1/1     Running   0          29s
nginx-rc-qj88f   1/1     Running   0          29s
nginx-rc-sgmhn   1/1     Running   0          29s
root@master:~#
```

**3 Nginx Pods running.**

**Task 4: Scale ReplicationController from 3 → 5**

```
root@master:~# kubectl scale rc nginx-rc --replicas=5
replicationcontroller/nginx-rc scaled
root@master:~# kubectl get rc
NAME       DESIRED   CURRENT   READY   AGE
nginx-rc   5         5         5       110s
root@master:~# kubectl get pods -l app=nginx-rc
NAME             READY   STATUS    RESTARTS   AGE
nginx-rc-4bt9c   1/1     Running   0          119s
nginx-rc-6jgs5   1/1     Running   0          18s
nginx-rc-ghr2b   1/1     Running   0          18s
nginx-rc-qj88f   1/1     Running   0          119s
nginx-rc-sgmhn   1/1     Running   0          119s
root@master:~#
```

**Task 5: ReplicaSet using multiple labels — `prod` and `test`**

**We'll create Pods with different labels and use a ReplicaSet selector that matches both prod and test.**

```
 GNU nano 8.7.1                                                rs-multiple-labels.yml *
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: nginx-rs
spec:
  replicas: 2
  selector:
    matchExpressions:
      - key: env
        operator: In
        values:
          - prod
          - test
  template:
    metadata:
      labels:
        app: nginx
        env: prod
    spec:
      containers:
        - name: nginx
          image: nginx
```

```
root@master:~# nano rs-multiple-labels.yml
root@master:~# kubectl create -f rs-multiple-labels.yml
replicaset.apps/nginx-rs created
root@master:~# kubectl get rs
NAME       DESIRED   CURRENT   READY   AGE
nginx-rs   2         2         2       7s
root@master:~# kubectl get pods --show-labels
NAME             READY   STATUS    RESTARTS   AGE     LABELS
apache-pod       1/1     Running   0          23m     app=apache
nginx-pod        1/1     Running   0          19m     app=nginx
nginx-rc-4bt9c   1/1     Running   0          3m49s   app=nginx-rc
nginx-rc-6jgs5   1/1     Running   0          2m8s    app=nginx-rc
nginx-rc-ghr2b   1/1     Running   0          2m8s    app=nginx-rc
nginx-rc-qj88f   1/1     Running   0          3m49s   app=nginx-rc
nginx-rc-sgmhn   1/1     Running   0          3m49s   app=nginx-rc
nginx-rs-5kptm   1/1     Running   0          16s     app=nginx,env=prod
nginx-rs-gfftf   1/1     Running   0          16s     app=nginx,env=prod
test-nginx       0/1     Error     0          7m3s    run=test-nginx
```

**we'll create a `test`-labelled Pod and demonstrate how the selector works.**

**Task 5 — verify the selector**

```
root@master:~# kubectl get rs nginx-rs -o yaml | grep -A10 matchExpressions
    matchExpressions:
    - key: env
      operator: In
      values:
      - prod
      - test
  template:
    metadata:
      labels:
        app: nginx
        env: prod
```

```
  GNU nano 8.7.1
apiVersion: v1
kind: Pod
metadata:
  name: test-label-pod
  labels:
    app: nginx
    env: test
spec:
  containers:
    - name: nginx
      image: nginx
```

```
root@master:~# nano test-label-pod.yml
root@master:~# kubectl create -f test-label-pod.yml
pod/test-label-pod created
root@master:~# kubectl get pods --show-labels
NAME             READY   STATUS    RESTARTS   AGE     LABELS
apache-pod       1/1     Running   0          26m     app=apache
nginx-pod        1/1     Running   0          21m     app=nginx
nginx-rc-4bt9c   1/1     Running   0          6m43s   app=nginx-rc
nginx-rc-6jgs5   1/1     Running   0          5m2s    app=nginx-rc
nginx-rc-ghr2b   1/1     Running   0          5m2s    app=nginx-rc
nginx-rc-qj88f   1/1     Running   0          6m43s   app=nginx-rc
nginx-rc-sgmhn   1/1     Running   0          6m43s   app=nginx-rc
nginx-rs-5kptm   1/1     Running   0          3m10s   app=nginx,env=prod
nginx-rs-gfftf   1/1     Running   0          3m10s   app=nginx,env=prod
test-nginx       0/1     Error     0          9m57s   run=test-nginx
root@master:~# kubectl get rs nginx-rs
NAME       DESIRED   CURRENT   READY   AGE
nginx-rs   2         2         2       3m20s
root@master:~# ^C
```

**First create the `test` Pod**

```
apiVersion: v1
kind: Pod
metadata:
  name: test-label-pod
  labels:
    app: nginx
    env: test
spec:
  containers:
    - name: nginx
      image: nginx
```

```
[{"key":"env","kubectl get pods -l 'env in (prod,test)' --show-labels# kubectl get pods -l 'env in (prod,test)' --show-labels
NAME           READY   STATUS    RESTARTS   AGE     LABELS
nginx-rs-5kptm   1/1     Running   0          7m46s   app=nginx,env=prod
nginx-rs-gfftf   1/1     Running   0          7m46s   app=nginx,env=prod
root@master:~#
```

**Task 6: ReplicaSet that excludes Pods with the backend label.**

```
 GNU nano 8.7.1                                                rs-exclude-backend.yml *
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: frontend-rs
spec:
  replicas: 2
  selector:
    matchExpressions:
      - key: tier
        operator: NotIn
        values:
          - backend
  template:
    metadata:
      labels:
        app: frontend
        tier: frontend
    spec:
      containers:
        - name: nginx
          image: nginx
```

```
root@master:~# nano rs-exclude-backend.yml
root@master:~# kubectl create -f rs-exclude-backend.yml
replicaset.apps/frontend-rs created
root@master:~# kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
frontend-rs   2         2         2       7s
nginx-rs      2         2         2       9m14s
root@master:~# ^C
```

**Task 7: Test load balancing across multiple Pods using NodePort**

```
 GNU nano 8.7.1                                                nginx-loadbalanc
apiVersion: v1
kind: Service
metadata:
  name: nginx-loadbalancer
spec:
  selector:
    app: nginx-rc
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30081
  type: NodePort
```

```
root@master:~# nano nginx-loadbalancer.yml
root@master:~# kubectl create -f nginx-loadbalancer.yml
service/nginx-loadbalancer created
root@master:~# kubectl get svc nginx-loadbalancer
NAME                 TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
nginx-loadbalancer   NodePort   10.96.116.116   <none>        80:30081/TCP   11s
root@master:~# kubectl get endpoints nginx-loadbalancer
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                 ENDPOINTS                                                  AGE
nginx-loadbalancer   192.168.171.11:80,192.168.171.12:80,192.168.37.201:80 + 2 more...   24s
root@master:~# ^C
root@master:~#
```

```
root@master:~# curl http://10.30.1.102:30081
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

**Kubernetes distributes incoming connections across the available backend Pods.**

**Task 8: Delete the ReplicationController without deleting its Pods.**

```
root@master:~# kubectl scale rc nginx-rc --replicas=5
replicationcontroller/nginx-rc scaled
root@master:~# kubectl get rc
NAME       DESIRED   CURRENT   READY   AGE
nginx-rc   5         5         5       34s
root@master:~# kubectl delete rc nginx-rc --cascade=orphan
replicationcontroller "nginx-rc" deleted from default namespace
root@master:~# kubectl get rc
kubectl get pods -l app=nginx-rc
No resources found in default namespace.
No resources found in default namespace.
root@master:~# ^C
```